Team Deuterium · PS-Q1 judges' report

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Plain-language explanation of what we built, how the 30 features become 4 qubits, how one heartbeat runs through the circuit, how well and how efficiently it works, and what we did not do. All numbers are validation results from executed runs (the test set has no labels for us).

One-paragraph summary. We classify ECG heartbeats as Normal or Arrhythmia with a 4-qubit quantum circuit. The 30 measured statistics of a beat are compressed into 4 numbers (principal components), each number becomes the rotation angle of one qubit, two trainable layers with a ring of CNOT gates mix the qubits, we measure qubit 0, and a trained scale and bias turn that measurement into a probability. The classical baseline (logistic regression) scores macro-F1 **1.000**; our circuit scores **0.9993** in ideal simulation (1 mistake in 1,500) and **0.466** under the organizers' noise model. We found why (noise shrinks the signal to 40% but keeps the ordering), showed a fix (0.9986), and made no claim of quantum advantage.

1. Why 30 features and only 4 qubits: how the 30 are reduced and used

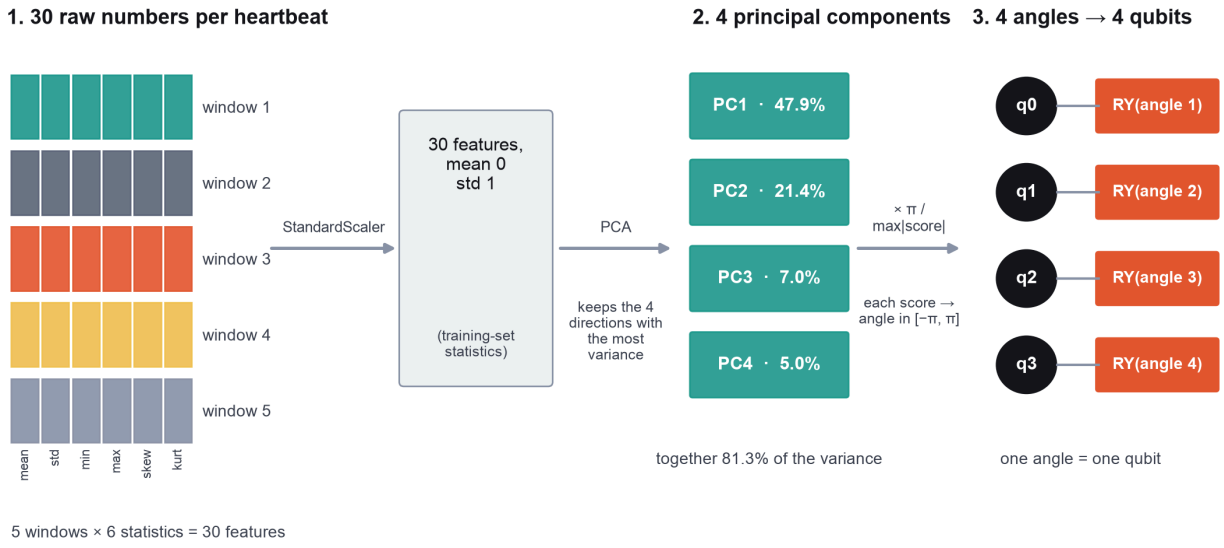


Figure 1. The whole reduction in one picture. Nothing is thrown away blindly: every one of the 30 numbers contributes to every principal component.

The question judges asked

"A circuit with 4 qubits cannot take 30 inputs, so how did you use all 30 features?" Answer: **each qubit receives one number, and each of those 4 numbers is a weighted mixture of all 30 features.** The mixing is learned automatically from the training data by Principal Component Analysis (PCA).

Step by step

1. **30 raw statistics.** A heartbeat is 187 samples at 360 Hz around the R-peak. It is cut into 5 windows of 35 samples; for each window we have mean, std, min, max, skewness and kurtosis: 5 × 6 = 30 numbers.
2. **StandardScaler.** The 30 numbers have different units and ranges (millivolts versus dimensionless skewness and kurtosis). Each feature is shifted to mean 0 and scaled to standard deviation 1 using *training-set* statistics only.
3. **PCA to 4.** The 30 features are strongly correlated (41% of all 435 feature pairs have |correlation| above 0.5, 8.5% above 0.8, Figure 2), so they carry much repeated information. PCA finds new axes (weighted sums of all 30 features) that capture the most variation first. The first 4 keep **81.3%** of the total variance (PC1 47.9%, 2 components 69.3%, 3 components 76.3%; 6 would keep 87.1%, 10 would keep 94.4%). Fitted on training data only.
4. **Angle scaling.** Each of the 4 scores is multiplied by π / (largest absolute training score) so it lies in [−π, π] (factors 0.366, 0.404, 0.480, 0.764). Without this, 52.8% of beats had an angle outside ±π and the circuit could not tell

them apart (Figure 6).

5. **One angle, one qubit.** Qubit i starts in |0⟩ and is rotated by RY(angle i). That is the only place the data enters the circuit.

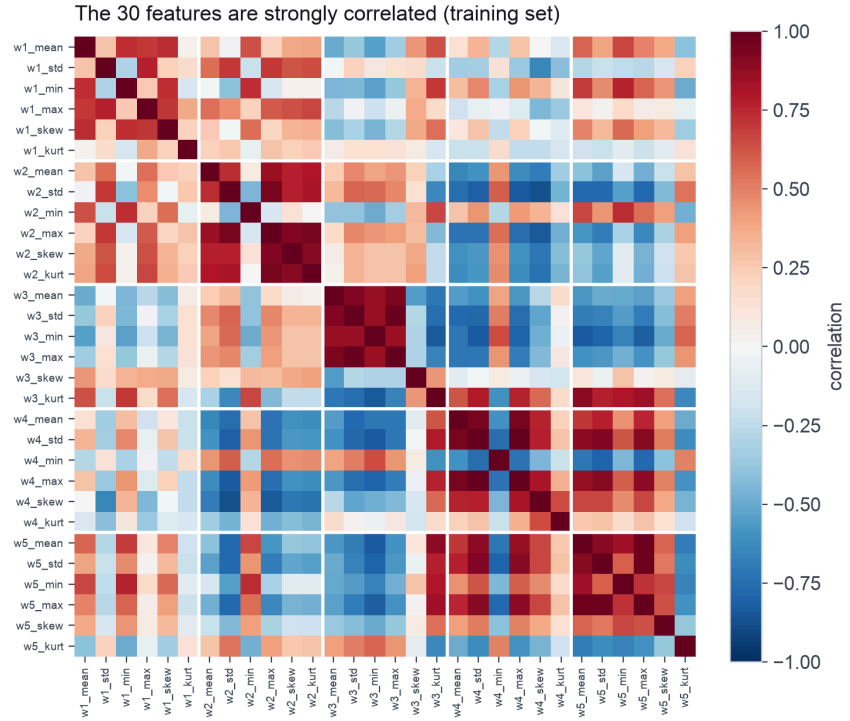


Figure 2. The 30 features are highly redundant (dark blocks), which is why compressing them loses little.

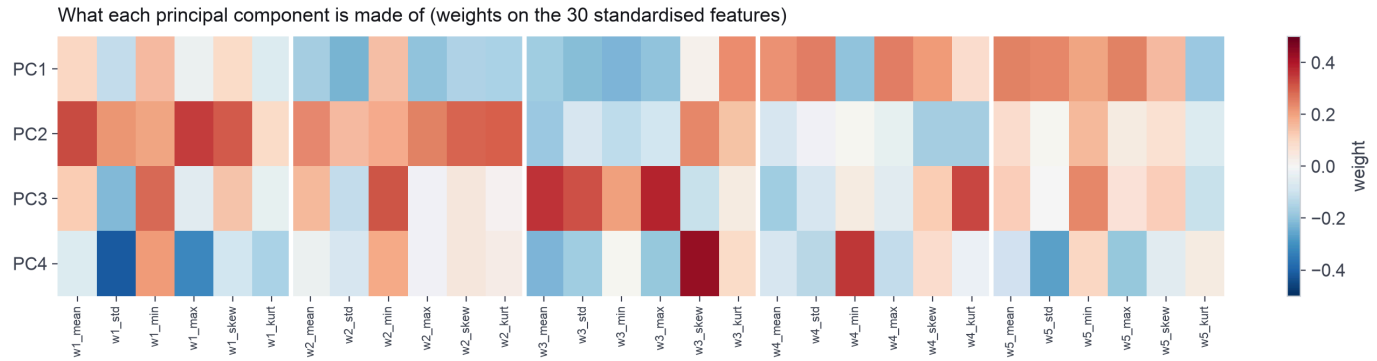


Figure 3. What each component is made of: every column is a feature, every row a component. All 30 features have non-zero weight in every component.

Component	Variance	Strongest features (weight)	Where it looks in the beat (share of weight by window)
PC1	47.9%	w4_std (+0.25), w4_max (+0.25), w5_mean (+0.25), w5_max (+0.25)	w1 6% · w2 19% · w3 22% · w4 27% · w5 27%: mostly the later part of the beat
PC2	21.4%	w1_max (+0.34), w1_mean (+0.33), w1_skew (+0.30), w2_kurt (+0.29)	w1 41% · w2 34% · w3 14% · w4 6% · w5 4%: mostly the early part
PC3	7.0%	w3_max (+0.38), w3_mean (+0.36), w4_kurt (+0.33), w3_std (+0.32)	w1 17% · w2 14% · w3 43% · w4 16% · w5 10%: the middle window
PC4	5.0%	w3_skew (+0.43), w1_std (-0.41), w4_min (+0.35), w1_max (-0.32)	w1 35% · w2 4% · w3 31% · w4 17% · w5 13%: early and middle windows

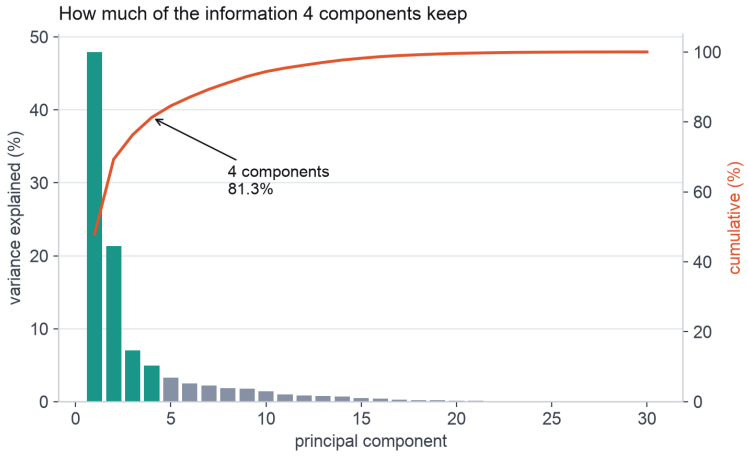


Figure 4. Variance explained by each of the 30 possible components; the 4 we keep are in teal.

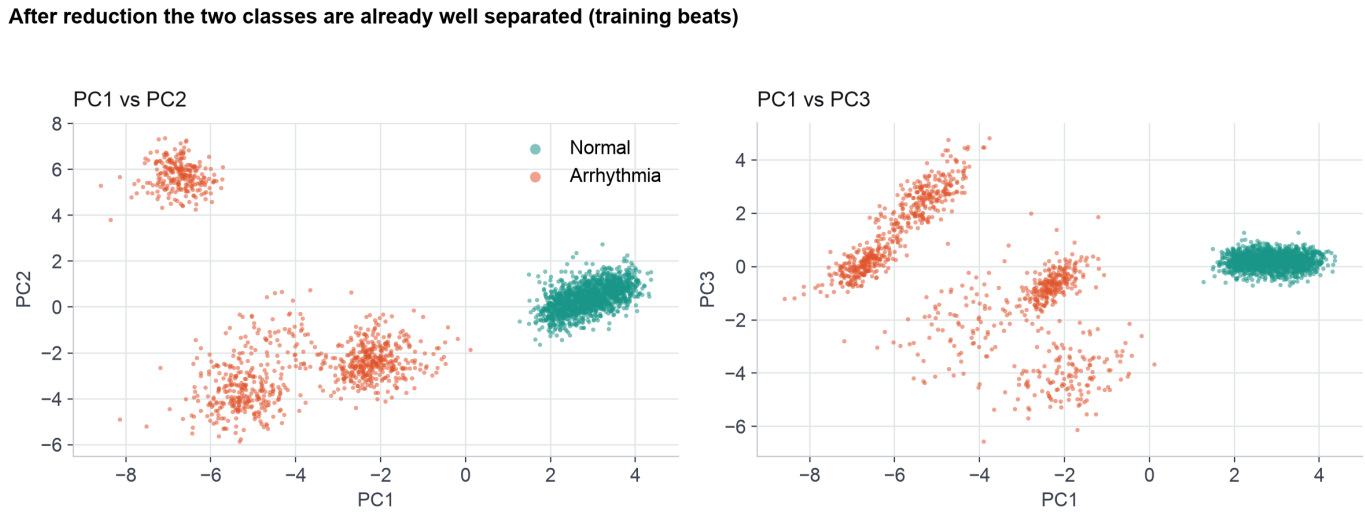


Figure 5. After reduction, Normal and Arrhythmia beats already separate; the quantum circuit only has to draw the boundary.

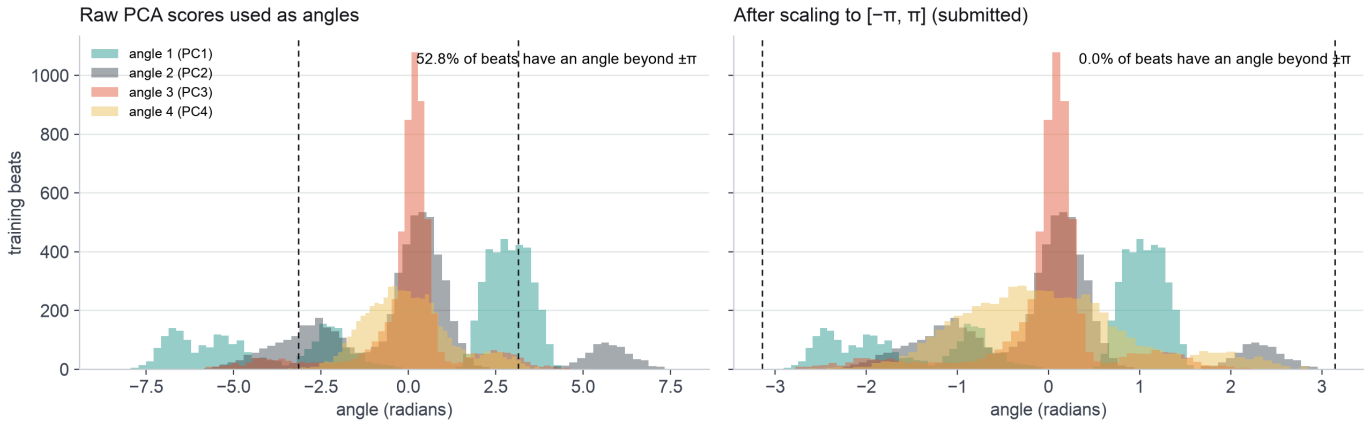


Figure 6. Why we rescale: raw scores (left) spill far beyond ±π and wrap around; after scaling (right) every beat fits in one turn.

Did the reduction lose useful information?

Checked directly: classical models given only the same 4 inputs score **macro-F1 1.0000** (logistic regression) and **1.0000** (RBF-SVM) on validation, the same as logistic regression on all 30 (1.0000). So the 4 numbers still contain what is needed; the remaining errors of the quantum model come from the circuit, not from the reduction. (These side experiments were run by `judges_pack/make_figures.py`, not in the notebook.)

Why 4, and what we did not test

- 4 qubits is inside the organizers' recommended 4 to 6 range; one qubit per component keeps the circuit small and the noisy simulation fast.
- We did **not** sweep 3, 5 or 6 components, and we did not try other ways of loading 30 numbers into fewer qubits (for example amplitude encoding, which can in principle hold 30 numbers in 5 qubits). These are listed in the gap

report (section 9).

- Using 30 qubits is not realistic: simulating n qubits costs about 2^n, and every extra qubit adds gates and noise.

2. How the quantum circuit runs, from one heartbeat to one decision

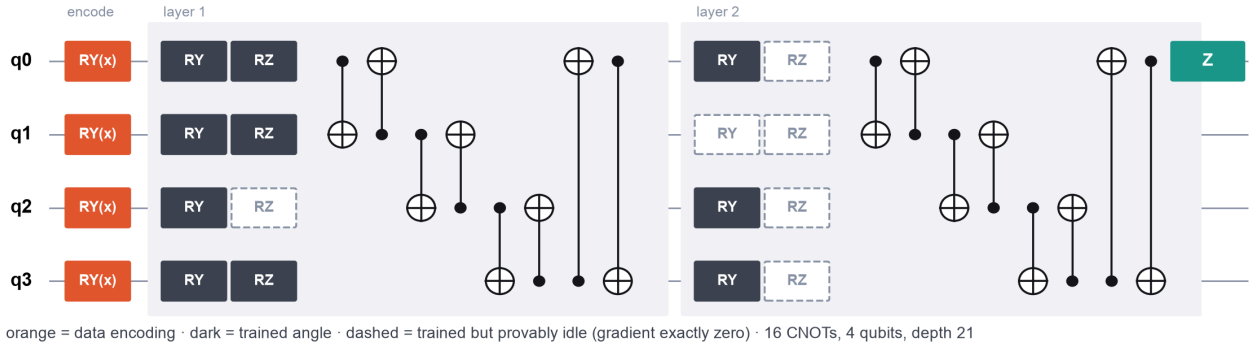


Figure 7. The circuit. Orange: data encoding. Dark: trained angles. Dashed: trained but provably idle. 4 qubits, 16 CNOT gates, depth 21.

The eight steps (inference for one beat)

1. **Input:** 30 numbers for one beat.
2. **Standardise** with the saved training means and standard deviations.
3. **Project to 4 components** with the saved PCA matrix.
4. **Scale to angles** with the saved factors.
5. **Encode:** apply RY(angle) to each of 4 fresh qubits. Each qubit's Z-value is now cos(angle).
6. **Process:** two layers; each layer applies a trained RY and a trained RZ to every qubit, then 8 CNOTs around the ring (0→1, 1→0, 1→2, 2→1, 2→3, 3→2, 3→0, 0→3). The CNOTs entangle the qubits so qubit 0 ends up depending on all four inputs.
7. **Measure qubit 0.** Ideal simulation computes the exact expected value $\langle Z_0 \rangle$ between −1 and +1. A noisy run repeats the circuit 8,192 times and uses (count of 0 − count of 1) / 8,192.
8. **Decide:** logit = 7.26 · $\langle Z_0 \rangle$ + 2.72, probability = sigmoid(logit), label Arrhythmia if probability ≥ 0.5 (that is $\langle Z_0 \rangle \geq -0.375$).

Training uses the same forward pass on batches of 16 training beats, computes the cross-entropy loss, gets exact gradients by backpropagation through the simulated state, and updates the 16 angles and the 2 readout numbers with Adam. Validation beats are only scored, never used for updates.

Three real beats from our validation set, end to end

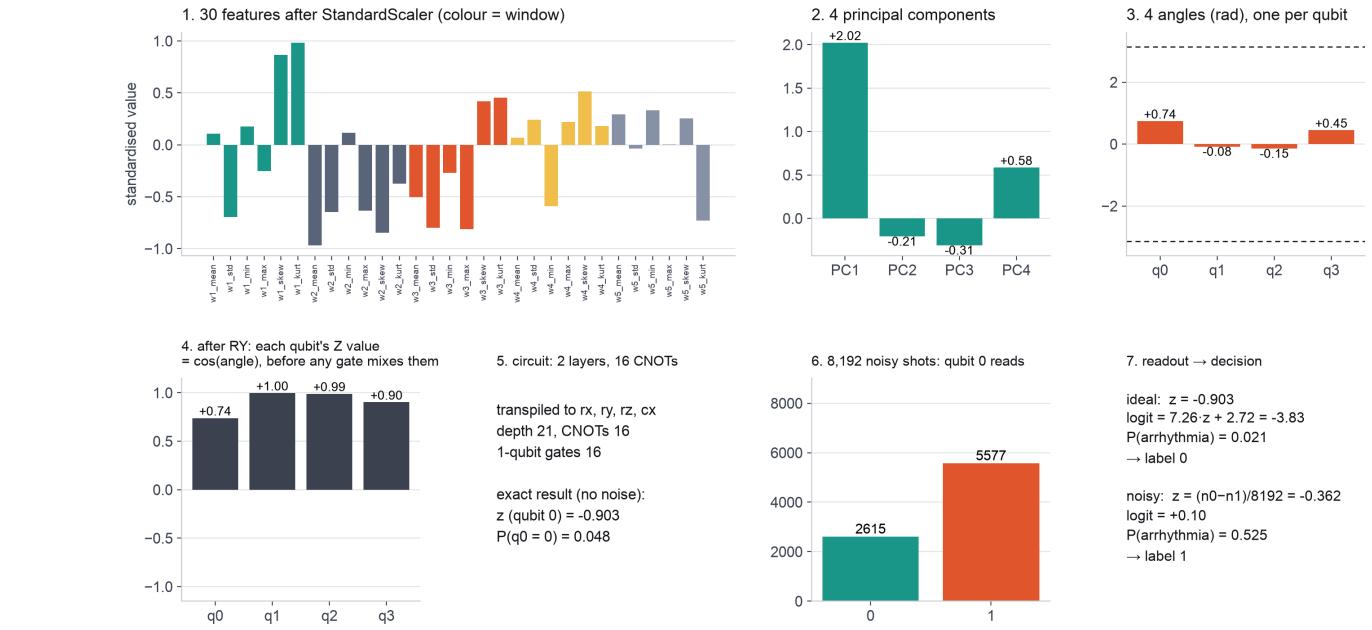
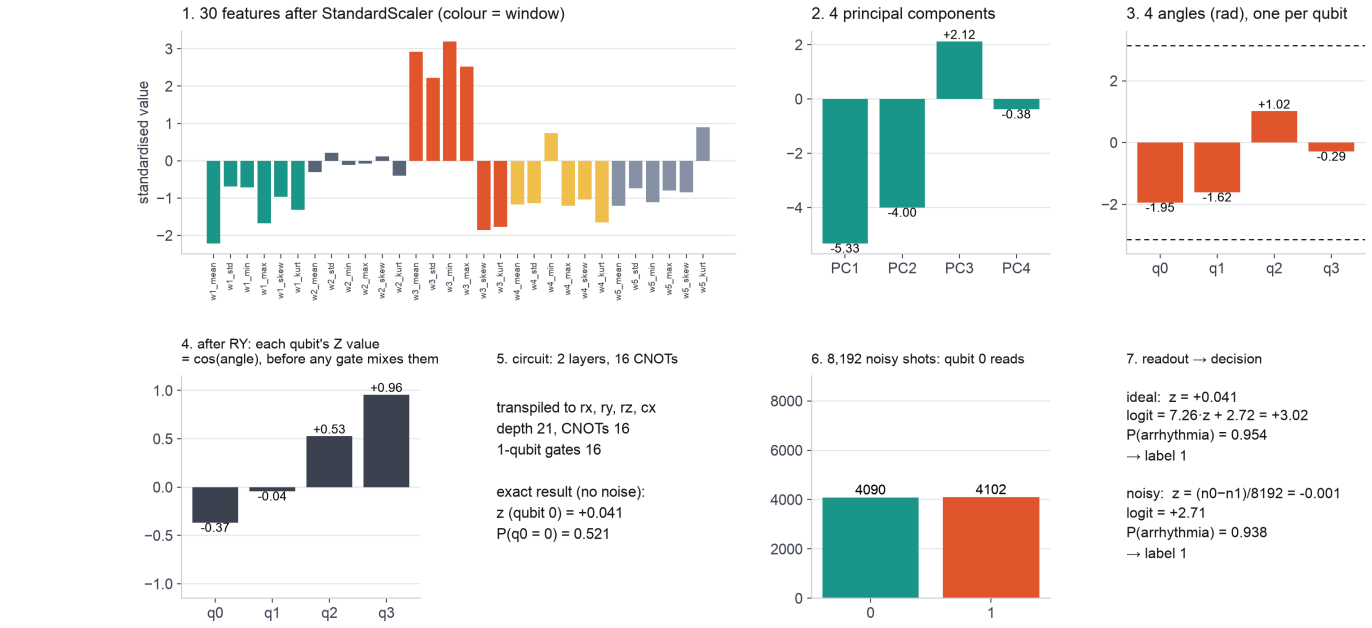


Figure 8. A Normal beat (validation row 0). Left to right, top to bottom: standardised features, principal components, angles, qubit values after encoding, circuit facts, noisy shot counts, readout and decision.



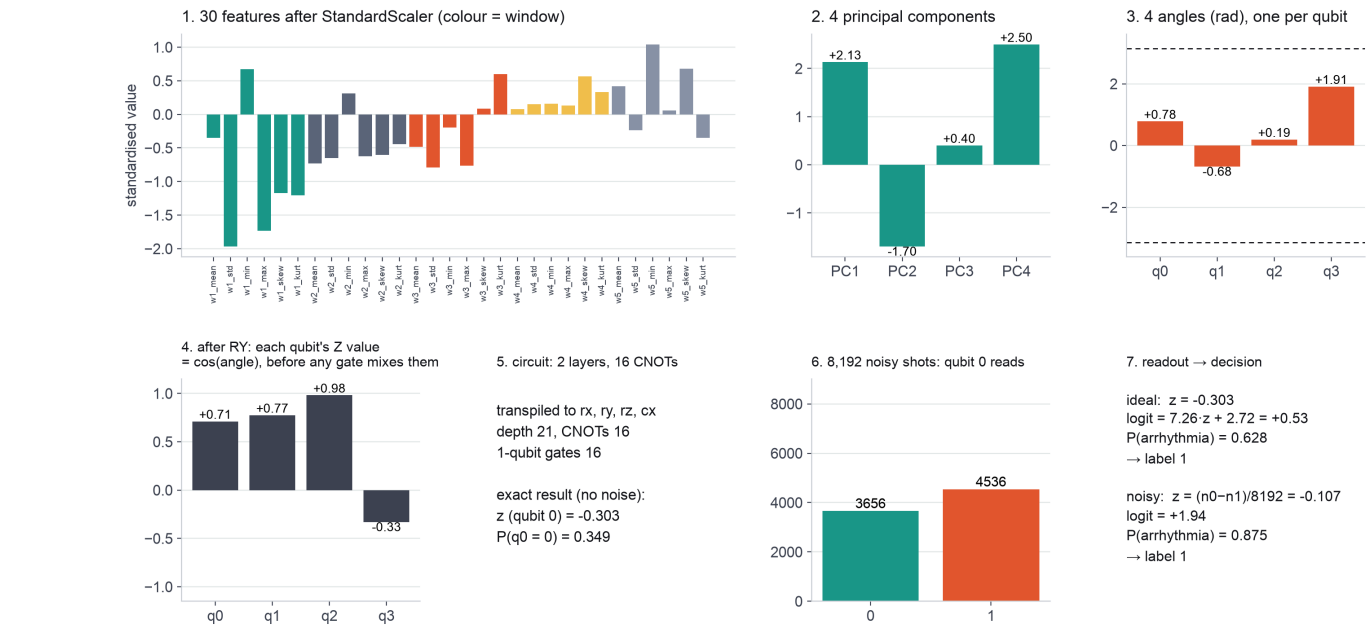


Figure 9. The only validation beat the ideal model gets wrong (a Normal beat called Arrhythmia with probability 0.628).

Beat	4 PCA scores	4 angles (rad)	ideal $\langle Z_0 \rangle$	ideal P(arr.) → label	noisy counts 0 / 1	noisy $\langle Z_0 \rangle$	noisy P(arr.) → label	truth
Normal beat (correct) row 0	+2.02, -0.21, -0.31, +0.58	+0.74, -0.08, -0.15, +0.45	-0.903	0.021 → 0	2615 / 5577	-0.362	0.525 → 1	Normal (0)
Arrhythmia beat (correct) row 8	-5.33, -4.00, +2.12, -0.38	-1.95, -1.62, +1.02, -0.29	+0.041	0.954 → 1	4090 / 4102	-0.001	0.938 → 1	Arrhythmia (1)
The one validation error row 664	+2.13, -1.70, +0.40, +2.50	+0.78, -0.68, +0.19, +1.91	-0.303	0.628 → 1	3656 / 4536	-0.107	0.875 → 1	Normal (0)

Every beat is transpiled to depth 21 with 16 CNOTs and 16 single-qubit gates (rx, ry, rz only). Notice row 0: ideally 0.021 (clearly Normal); under noise the signal shrinks toward zero and the probability moves to 0.525, which flips the label. That is exactly the failure analysed in section 5.

3. Training and validation loss

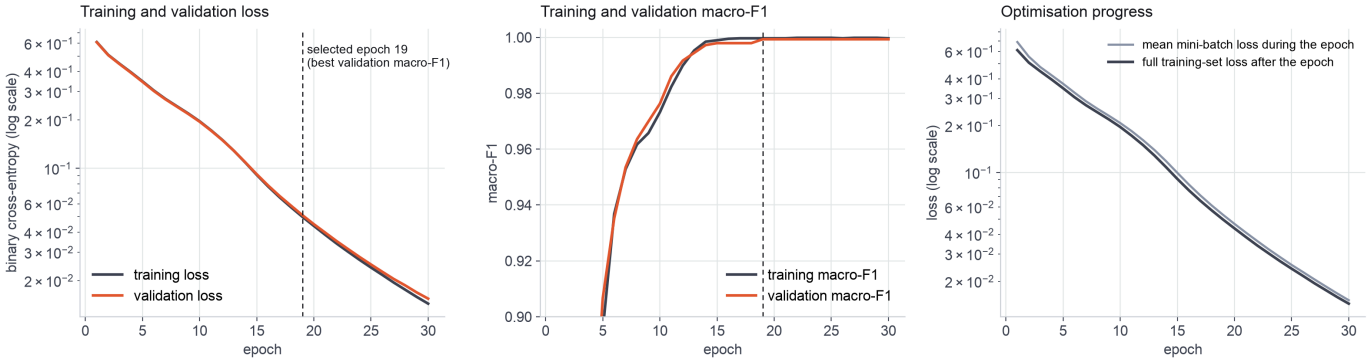


Figure 10. Submitted model (scaled angles, 2 layers, learning rate 0.001).

- **Loss falls smoothly:** training loss 0.612 after epoch 1 to 0.0144 after epoch 30; validation loss 0.609 to 0.0154.
- **No overfitting:** the two curves lie on top of each other the whole way (at the selected epoch 19: training 0.0499, validation 0.0511; training macro-F1 0.9997, validation 0.9993). A model with 18 parameters on 6,000 beats has little room to memorise.
- **Why epoch 19:** macro-F1 reaches its best value (0.9993) at epoch 19 and stays there; the rule picks the earliest best epoch. Loss keeps falling afterwards, but the decisions do not change.
- The right panel shows the optimiser is stable: mini-batch loss tracks the full training loss closely.

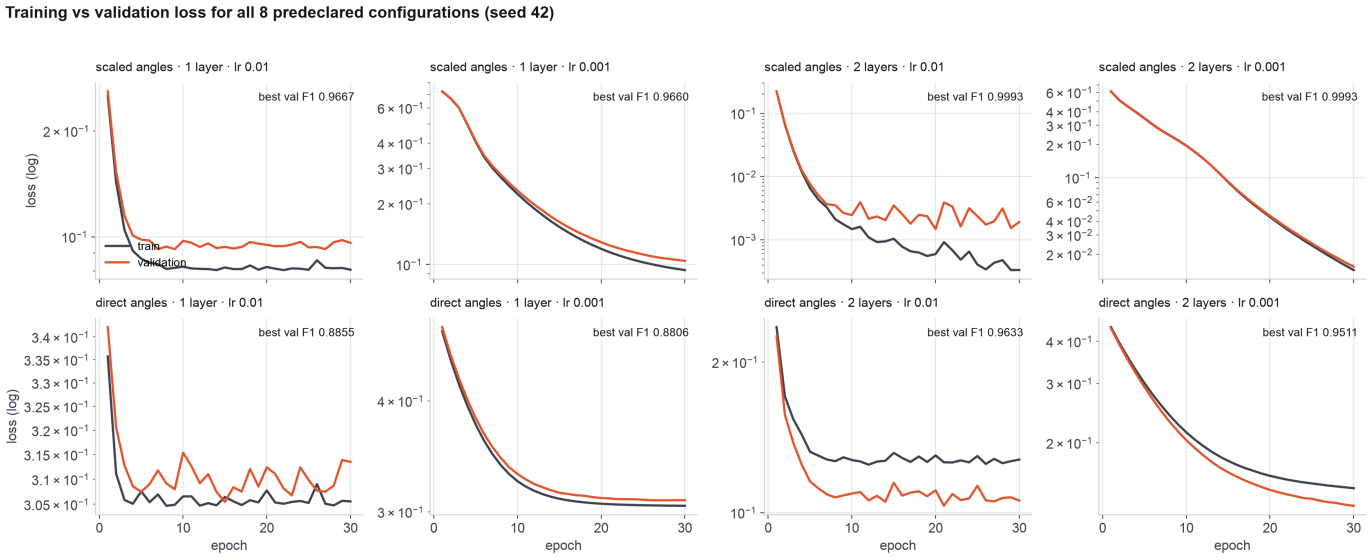


Figure 11. All 8 predeclared configurations. 2-layer models with scaled angles reach the lowest loss; direct (unscaled) angles plateau at a much higher loss, which is the angle-wrapping problem showing up in the training curves.

Configuration	final train loss	final val loss	best val macro-F1
Scaled, 1 layer, lr 0.01	0.0806	0.0961	0.9667
Scaled, 1 layer, lr 0.001	0.0938	0.1042	0.9660
Scaled, 2 layers, lr 0.01	0.0003	0.0019	0.9993
Scaled, 2 layers, lr 0.001 (submitted)	0.0144	0.0154	0.9993
Direct, 1 layer, lr 0.01 / 0.001	0.3056 / 0.3048	0.3135 / 0.3092	0.8855 / 0.8806
Direct, 2 layers, lr 0.01 / 0.001	0.1276 / 0.1467	0.1057 / 0.1301	0.9633 / 0.9511

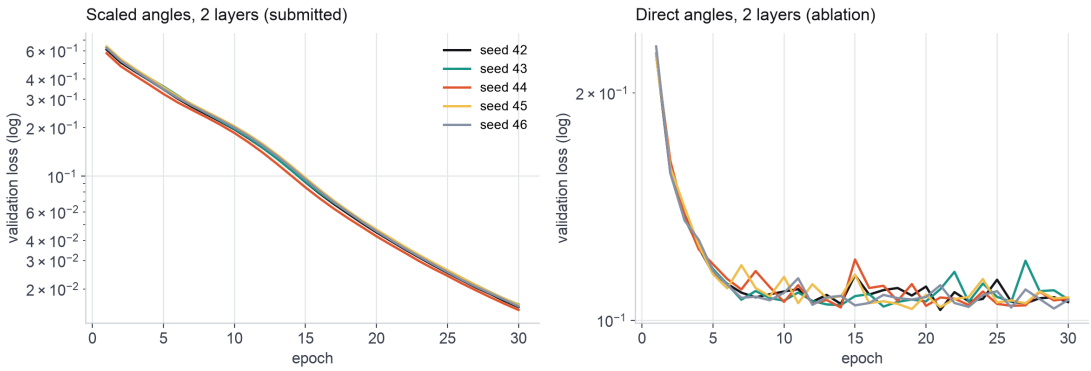


Figure 12. Five training seeds give nearly the same validation-loss curve, so the result does not depend on a lucky start.

4. What the models do, and how good they are

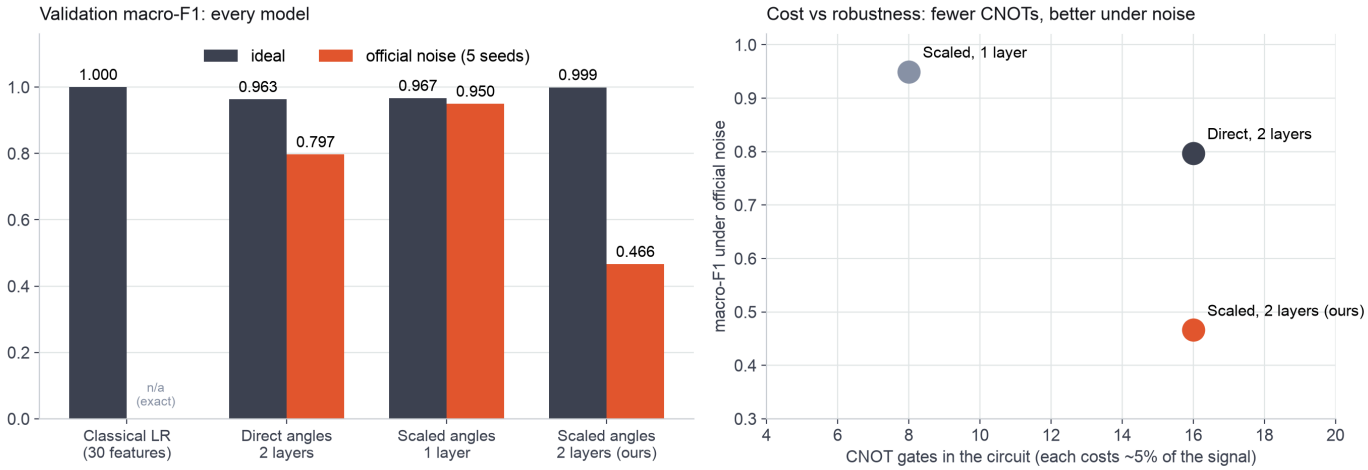


Figure 13. Left: validation macro-F1 of every model, ideal and under the official noise model. Right: the more CNOT gates, the more signal is lost under noise.

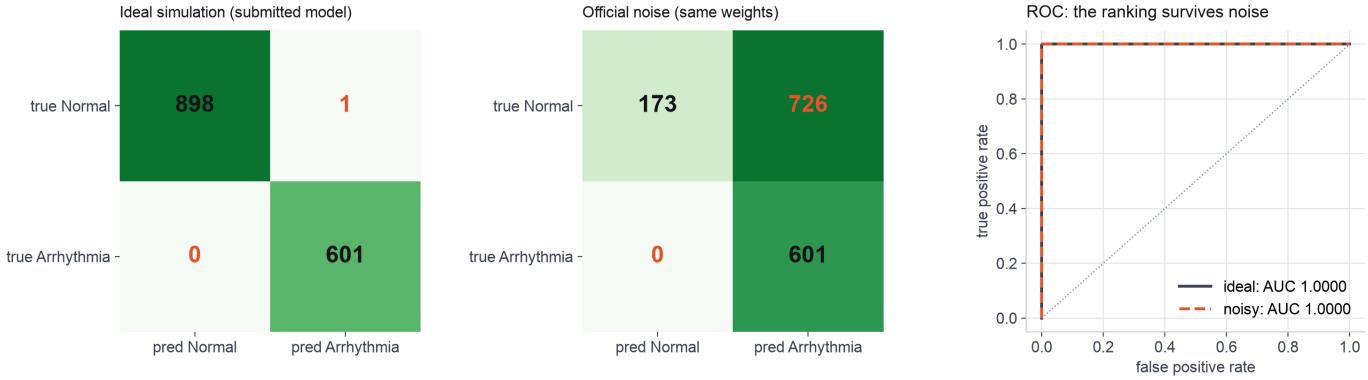


Figure 14. Submitted model. Ideal: 898 Normal and 601 Arrhythmia correct, 1 mistake. Under noise: 726 Normal beats are wrongly called Arrhythmia, none missed. The ROC curves overlap (AUC 1.0000 both): the model still ranks beats correctly, only the cut-off moves.

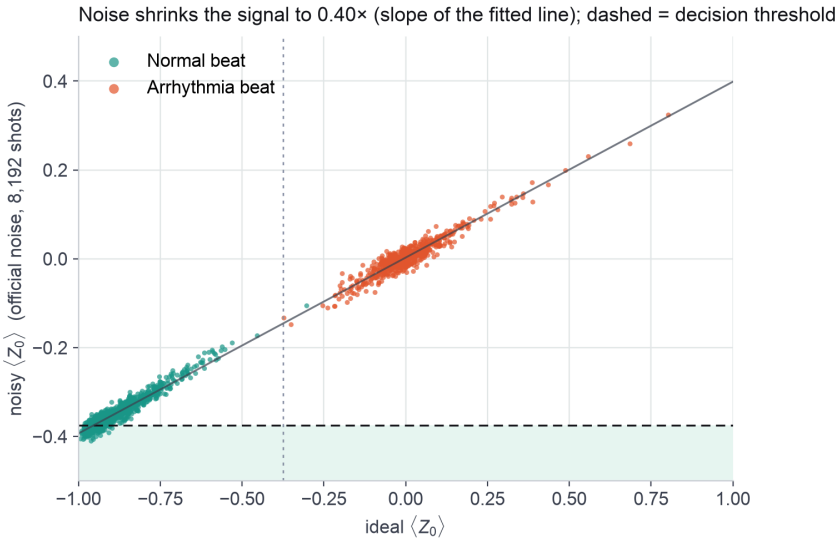


Figure 15. Every dot is one validation beat. Noisy $\langle Z_0 \rangle$ is the ideal value times 0.40 (correlation 0.9977); the learned threshold (dashed) was set for full-strength signals.

5. Differences between all the models

	Logistic regression (classical)	Direct angles, 2 layers	Scaled, 1 layer	Scaled, 2 layers (submitted)
What it does	weighted sum of 30 features → probability	quantum circuit on raw PCA angles	one layer, 8 CNOTs	two layers, 16 CNOTs
Inputs	30 standardised features	4 PCA scores (unscaled)	4 scaled angles	4 scaled angles
Trainable numbers	31	18 (12 effective)	10 (5 effective)	18 (12 effective)
Ideal macro-F1 (95% CI)	1.0000	0.9633 (0.953–0.973)	0.9667 (0.957–0.975)	**0.9993** (0.998–1.000)
Under official noise	not applicable (exact)	0.7971 ± 0.0037	**0.9496** ± 0.0013	0.4663 ± 0.0048
Signal kept under noise	n/a	0.38×	0.60×	0.40×
Sees all 4 inputs?	yes (all 30)	yes	no: ignores PC2	yes
Main weakness	none on this data	angles wrap around	blind to PC2, lower accuracy	fragile under noise
Role in the project	baseline	ablation (encoding)	ablation (depth)	submitted model

Other comparisons

Comparison	Result
Classical models on the **same 4 inputs** (validation macro-F1)	logistic regression 1.0000, RBF-SVM 1.0000: classical models handle these inputs perfectly, so the quantum circuit is the limiting piece, not the data
Ideal vs noisy simulation	ideal = exact expectation values (PennyLane); noisy = 8,192 shots on Qiskit-Aer with the official QFF2026-v2 model (1% / 5% / 2%) after transpiling to rx, ry, rz, cx
Phase 1 vs Phase 2 *(Phase 1 from its repo, not re-run)*	Phase 1: synthetic data (200 samples, 4 features), H/RZ feature map, ZZZZ-parity readout, COBYLA, own weak noise, accuracy 0.925 on 40 samples. Phase 2: real ECG statistics (30 features, 9,000 labelled beats), PCA + angle scaling, single-qubit readout with trained scale and bias, Adam, official noise, macro-F1 0.9993 on 1,500 beats
Processor designs (Task B, noisy QFT-3 fidelity)	linear one-way 0.550 · linear two-way 0.604 · ring 0.635 · ring + chord 0.668 · ring + chord + native H **0.6775** (submitted, 0 SWAPs)

6. How efficient are the models

Cost	Logistic regression	VQC 1 layer	VQC 2 layers (submitted)
Trainable numbers	31	10 (5 effective)	18 (12 effective)
Qubits / depth / CNOTs	n/a	4 / 11 / 8	4 / 21 / 16
Training time	12 ms (11 solver iterations)	about 155 s	about 225 s (30 epochs; 8 runs shared a 10-core CPU)
Circuit evaluations in training	none	405,000 forward passes	405,000 forward passes (gradients by backprop: no extra circuit runs)
Inference on 1,500 beats	0.6 ms	about 12 ms (exact simulation)	about 20 ms (exact); about 8.7 s per 8,192-shot noisy pass
Noisy evaluation cost	none	61.44 M shots (1,500 circuits × 8,192 × 5 seeds)	61.44 M shots
Accuracy per CNOT (noisy macro-F1 / CNOTs)	n/a	**0.119**	0.029

Honest reading. The classical model is cheaper in every way and perfect on this data. The quantum model's efficiency story is internal: 18 numbers and 4 qubits are enough to reach 0.9993 ideally; every CNOT costs about 5% of the signal, so halving the CNOTs triples the noisy score per gate. We do not claim quantum advantage.

7. Why this project stands out

- **We explain, not just report.** Each result has a measured cause: angle wrapping (+3.6 points), idle parameters found with a Heisenberg-picture argument, signal shrinkage (0.40×) with a preserved ranking, and a readout recalibration that recovers 0.9986.
- **Honest comparison.** We put the classical 1.000 on the first content slide, show seeds and confidence intervals, and state that we do not beat it.
- **Verification culture.** 12 automated checks pass; Qiskit and PennyLane circuits agree to 1e-15 before and after transpilation; noise is proven applied (readout flip measured at 1.81% against 2%); we found and fixed an evaluation shortcut of our own; two full runs gave identical outputs; notebook runs from a fresh environment in about 9.5 minutes.
- **Hardware-circuit co-design.** Our processor is a 4-qubit ring plus one chord: our classifier runs there with 16 CNOTs instead of 34 on the linear example, and the processor scores 0.6775 on the official QFT-3 benchmark with no SWAPs.
- **Rules followed strictly.** Everything learned uses training data only; the test file is opened once for inference; kit files match the organizers' checksums; all noisy runs use the official helper unchanged.

Rubric category	Where we show it
C1 Quantum design (30)	sections 1, 2, 5, 6; ablations (encoding, depth); resource table; idle-parameter analysis
C2 Results (25)	section 4 and 5: classical baseline, seeds, intervals, honest comparison; predictions validated
C3 Noise (20)	Task A (F 0.8801), ideal vs noisy for every model, why it drops, mitigation
C4 Processor D (15)	five designs measured; directed links, SWAPs, depth explained; 16 vs 34 CNOTs
C5 Reproducibility (10)	seeds, pip freeze, self-test, identical reruns, fresh-environment run, full report

8. Task A and Task B (short)

Task A (8,192 shots)	⟨XX⟩	⟨YY⟩	⟨ZZ⟩	F
Ideal	+1.0000	−1.0000	+1.0000	1.0000
Noisy	+0.8191	−0.8279	+0.8735	**0.8801 ± 0.0026**

Noise leaves the entanglement intact (F far above the 0.5 of a classical mixture) but removes about 12% of the overlap after a single CNOT, which is why a 16-CNOT classifier suffers so much. Processor D: ring + chord 0-2, all links two-way, gates rx ry rz h cx; official benchmark depth 16, 9 two-qubit gates, 0 SWAPs, noisy fidelity 0.677499 (ideal 1.0).

9. Gap report: what we have not done or could have missed

Judges may press on these. Say them first.

#	Gap	Why it matters	Status / what to say
1	Loss curves, ROC and confusion plots were missing from the notebook and results.md	judges asked for them	now produced in `judges_pack/figures`; **not yet in the GitHub notebook**
2	Number of principal components never swept (3, 5, 6)	4 is justified but not shown optimal	"we chose 4 for qubit count; classical models on those 4 are perfect; sweep is future work"
3	No alternative encodings (amplitude encoding, data re-uploading)	our model class is limited (81 terms)	listed as next step
4	Ring direction, observable, qubit count, depth 3, class weights untested	design choices not ablated	"untested; first experiments we would run"
5	6 of 16 angles idle	wasted gates and noise	documented; pruning untested
6	Submitted model collapses under noise (0.466)	weakest result	explained and mitigated; 1-layer is robust (0.950)
7	Validation chose the model; no held-out score; encoding change made after seeing validation	optimistic numbers	disclosed everywhere
8	Synthetic, easy data (one feature separates it); no real ECG; no patient-wise split	no clinical claim possible	state clearly
9	No hardware run; official noise is an idealised model	deployment claims	make none
10	Test-set angle range never inspected	unseen beats could wrap	validation had 0% outside ±π
11	No formal significance test vs classical (differs by 1 sample)	statistics	intervals and seeds only
12	Full 8-run search not repeated; Phase 1 not re-run	reproducibility	only the selected model was retrained twice
13	Teammate's classical reports contain file paths from their Windows machine	looks unpolished	content is valid; mention paths are theirs
14	Repo is private; second repo `Qiskit-VQC-CN` exists	portal needs a public repo	make `Qiskit-VQC-CNn` public; make other private
15	Team roles and pitch length undocumented	pitch preparation	confirm before presenting

10. Where every report is

What	File
This explanatory report	`judges_pack/01_Judges_Report.pdf`

All written reports combined in one PDF	`judges_pack/02_All_Reports_Combined.pdf`
Every figure (16 images)	`judges_pack/figures/`
Individual report files, data artifacts, notebook, predictions	`judges_pack/reports/`, `judges_pack/artifacts/`, `judges_pack/notebook/`
Slide deck and pitch material	`judges_pack/presentation/`
Code that reproduces the figures	`judges_pack/make_figures.py`